

41.0.1 UGC NET CSE | November 2017 | Part 2 | Question: 2



Let $m = (313)_4$ and $n = (322)_4$. Find the base 4 expansion of $m + n$.

- A. $(635)_4$ B. $(32312)_4$ C. $(21323)_4$ D. $(1301)_4$

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Answer key

41.0.2 UGC NET CSE | November 2017 | Part 2 | Question: 4



How many distinguishable permutations of the letters in the word of BANANA are there?

- A. 720 B. 120 C. 60 D. 360

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Answer key

41.0.3 UGC NET CSE | November 2017 | Part 2 | Question: 3



Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ Find the boolean product $A \odot B$ of the two matrices.

A. $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

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41.0.4 UGC NET CSE | November 2017 | Part 2 | Question: 9



Negation of the proposition $\exists x H(x)$ is

- A. $\exists x \neg H(x)$ B. $\forall x \neg H(x)$
C. $\forall x H(x)$ D. $\neg x H(x)$

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Answer key 

41.0.5 UGC NET CSE | November 2017 | Part 2 | Question: 11



prtdata is a pointer to a data type. The expression *prtdata ++ is evaluated as (in C++)

- A. *(prtdata++)
- B. (*prtdata)++
- C. *(prtdata)++
- D. Depends on compiler

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Answer key 

41.0.6 UGC NET CSE | November 2017 | Part 2 | Question: 12



The associativity of which of the following operators is Left to Right, in C++?

- A. Unary Operator
- B. Logical not
- C. Array element access
- D. addressof

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Answer key 

41.0.7 UGC NET CSE | November 2017 | Part 2 | Question: 13



A member function can always access the data in, (in C++)

- A. the class of which it is member
- B. the object of which it is a member
- C. the public part of its class
- D. the private part of its class

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Answer key 

41.0.8 UGC NET CSE | November 2017 | Part 2 | Question: 14



Which of the following is **not** correct for virtual function in C++?

- A. Must be declared in public section of class
- B. Virtual function can be static
- C. Virtual function should be accessed using pointers
- D. Virtual function is defined in base class

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Answer key 

41.0.9 UGC NET CSE | November 2017 | Part 2 | Question: 16



Which of the following is/are **true** with reference to 'view' in DBMS?

- i. A 'view' is a special stored procedure executed when certain event occurs
- ii. A 'view' is a virtual table, which occurs after executing a pre-compiled query

- A. Only (i) is true
- B. Only (ii) is true
- C. Both (i) and (ii) are true
- D. Neither (i) nor (ii) are true

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Answer key 

41.0.10 UGC NET CSE | November 2017 | Part 2 | Question: 15



Which of the following is **not** correct (in C++)?

- A. Class templates and function templates are instantiated in the same way
- B. Class templates differ from function templates in the way they are initiated
- C. Class template is initiated by defining an object using the template argument
- D. Class templates are generally used for storage classes

Answer key

41.0.11 UGC NET CSE | November 2017 | Part 2 | Question: 21

Consider an array representation of an n element binary heap where the elements are stored from index 1 to index n of the array. For the element stored at index i of the array ($i \leq n$), the index of the parent is

- A. $\text{floor}((i+1)/2)$ B. $\text{ceiling}((i+1)/2)$
C. $\text{floor}(i/2)$ D. $\text{ceiling}(i/2)$

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Answer key 

41.0.12 UGC NET CSE | November 2017 | Part 2 | Question: 19

In RDBMS, different classes of relations are created using _____ technique to prevent modification anomalies.

- A. Functional Dependencies
B. Data integrity
C. Referential integrity
D. Normal Forms

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Answer key

41.0.13 UGC NET CSE | November 2017 | Part 2 | Question: 20

SQL command changes one or more fields in a record

- A. LOOK-UP B. INSERT C. MODIFY D. CHANGE

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Answer key

41.0.14 UGC NET CSE | November 2017 | Part 2 | Question: 18

Match the following with respect to RDBMS :

- | | |
|---------------------------|---|
| (a) Entity integrity | (i) enforces some specific business rule that do not fall into entity or domain |
| (b) Domain integrity | (ii) Rows can't be deleted which are used by other records |
| (c) Referential integrity | (iii) enforces valid entries for a column |
| (d) Userdefined integrity | (iv) No duplicate rows in a table |

Code :

- A. (a)-(iii); (b)-(iv); (c)-(i); (d)-(ii)
 B. (a)-(iv); (b)-(iii); (c)-(ii); (d)-(i)
 C. (a)-(iv); (b)-(ii); (c)-(iii); (d)-(i)
 D. (a)-(ii); (b)-(iii); (c)-(iv); (d)-(i)

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Answer key

41.0.15 UGC NET CSE | November 2017 | Part 2 | Question: 22

The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree?

- A. 3 B. 4 C. 5 D. 6

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Answer key 

41.0.16 UGC NET CSE | November 2017 | Part 2 | Question: 25

Post-order traversal of a given binary search tree T produces following sequence of keys: 3, 5, 7, 9, 4, 17, 16, 20, 18, 15, 14. Which one of the following sequences of keys can be the result of an in-order traversal of the tree T ?

- A. 3, 4, 5, 7, 9, 14, 20, 18, 17, 16, 15
B. 20, 18, 17, 16, 15, 14, 3, 4, 5, 7, 9
C. 20, 18, 17, 16, 15, 14, 9, 7, 5, 4, 3
D. 3, 4, 5, 7, 9, 14, 15, 16, 17, 18, 20

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Answer key

41.0.17 UGC NET CSE | November 2017 | Part 2 | Question: 26

Which of the following devices takes data sent from one network device and forwards it to the destination node based on MAC address?

- A. Hub B. Modem C. Switch D. Gateway

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Answer key

41.0.18 UGC NET CSE | November 2017 | Part 2 | Question: 27

_____ do not take their decisions on measurements or estimates of the current traffic and topology.

- A. Static algorithms B. Adaptive algorithms
C. Non-Adaptive algorithms D. Recursive algorithms.

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Answer key

41.0.19 UGC NET CSE | November 2017 | Part 2 | Question: 23

Let G be an undirected connected graph with distinct edge weight. Let $E_{\max}$ be the edge with maximum weight and $E_{\min}$ the edge with minimum weight. Which of the following statements is false?

- A. Every minimum spanning tree of G must contain $E_{\min}$
B. If $E_{\max}$ is in minimum spanning tree, then its removal must disconnect G
C. No minimum spanning tree contains $E_{\max}$
D. G has a unique minimum spanning tree

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Answer key

41.0.20 UGC NET CSE | November 2017 | Part 2 | Question: 24

A list of n strings, each of length n , is sorted into lexicographic order using merge - sort algorithm. The worst case running time of this computation is:

- A. $O(n \log n)$ B. $O(n^2 \log n)$ C. $O(n^2 + \log n)$ D. $O(n^3)$

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41.0.21 UGC NET CSE | November 2017 | Part 2 | Question: 36

In Distributed system, the capacity of a system to adapt the increased service load is called

- A. Tolerance B. Scalability C. Capability D. Loading

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Answer key

41.0.22 UGC NET CSE | November 2017 | Part 2 | Question: 39



Which module gives control of the CPU to the process selected by the short-term scheduler?

- A. Dispatcher B. Interrupt C. Scheduler D. Threading

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Answer key

41.0.23 UGC NET CSE | November 2017 | Part 2 | Question: 41



Software does not wear-out in the traditional sense of the term, but software does tend to deteriorate as it evolves, because

- A. Software suffers from exposure to hostile environments
B. Defects are more likely to arise after software has been used often
C. Multiple change requests introduce errors in component interactions
D. Software spare parts become harder to order

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Answer key

41.0.24 UGC NET CSE | November 2017 | Part 2 | Question: 40



Two atomic operations permissible on Semaphores are _____ and _____.

- A. wait, stop B. wait, hold C. hold, signal D. wait, signal

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Answer key

41.0.25 UGC NET CSE | November 2017 | Part 2 | Question: 42



Software re-engineering is concerned with

- A. Re-constructing the original source code from the existing machine (low-level) code program and modifying it to make it more user-friendly
B. Scrapping the source code of a software and re-writing it entirely from scratch
C. Re-organising and modifying existing software systems to make them more maintainable
D. Translating source code of an existing software to a new machine (low-level) language

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Answer key

41.0.26 UGC NET CSE | November 2017 | Part 2 | Question: 38



Suppose there are six files $F_1, F_2, F_3, F_4, F_5, F_6$ with corresponding sizes 150 KB, 225 KB, 75 KB, 60 KB, 275 KB and 65 KB respectively. The files are to be stored on a sequential device in such a way that optimizes access time. In what order should the files be stored?

- A. $F_5, F_2, F_1, F_3, F_6, F_4$ B. $F_4, F_6, F_3, F_1, F_2, F_5$
C. $F_1, F_2, F_3, F_4, F_5, F_6$ D. $F_6, F_5, F_4, F_3, F_2, F_1$

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Answer key

41.0.27 UGC NET CSE | November 2017 | Part 2 | Question: 43



Which of the following is **not** a key issue stressed by an agile philosophy of software engineering?

- A. The importance of self-organizing teams as well as communication and collaboration between team members and customers

- B. Recognition that change represents opportunity
- C. Emphasis on rapid delivery of software that satisfies the customer
- D. Having a separate testing phase after a build phase

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Answer key 

41.0.28 UGC NET CSE | November 2017 | Part 2 | Question: 44



What is the normal order of activities in which traditional software testing is organized?

- | | |
|------------------------|-----------------------|
| a. Integration Testing | b. System Testing |
| c. Unit Testing | d. Validation Testing |
| A. (c), (a), (b), (d) | B. (c), (a), (d), (b) |
| C. (d), (c), (b), (a) | D. (b), (d), (a), (c) |

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Answer key 

41.0.29 UGC NET CSE | November 2017 | Part 2 | Question: 49



Which of the given wireless technologies used in IoT, consumes the least amount of power?

- | | | | |
|-----------|--------------|----------|-------------|
| A. Zigbee | B. Bluetooth | C. Wi-Fi | D. GSM/CDMA |
|-----------|--------------|----------|-------------|

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Answer key 

41.0.30 UGC NET CSE | November 2017 | Part 3 | Question: 16



Find the equation of the circle $x^2 + y^2 = 1$ in terms of $x'y'$ coordinates, assuming that the xy coordinate system results from a scaling of 3 units in the x' direction and 4 units in the y' direction.

- | | |
|----------------------------|--|
| A. $3(x')^2 + 4(y')^2 = 1$ | B. $\left(\frac{x'}{3}\right)^2 + \left(\frac{y'}{4}\right)^2 = 1$ |
| C. $(3x')^2 + (4y')^2 = 1$ | D. $\frac{1}{3}(x')^2 + \frac{1}{4}(y')^2 = 1$ |

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Answer key 

41.0.31 UGC NET CSE | November 2017 | Part 3 | Question: 17



Find the normalization transformation that maps a window whose lower left corner is at $(1, 1)$ and upper right corner is at $(3, 5)$ onto a viewpoint that is the entire normalized device screen

A.
$$\begin{pmatrix} \frac{1}{2} & 0 & \frac{-1}{2} \\ 0 & \frac{1}{4} & \frac{-1}{4} \\ 0 & 0 & 1 \end{pmatrix}$$

B.
$$\begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{-1}{4} & \frac{1}{4} \\ 1 & 1 & 1 \end{pmatrix}$$

C.

$$\text{D. } \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{4} & \frac{-1}{4} \\ 1 & 0 & 0 \end{pmatrix}$$

Answer key 

A. $\Omega(n)$

$$\text{B. } \Omega\left(\frac{n}{k}\right)$$

$$\text{D. } \Omega\left(\frac{n}{k} \lg \frac{n}{k}\right)$$

Answer key

A. n

C. $n \lg n$

Answer key

A. 5

C. 4

Answer key 

41.0.35 UGC NET CSE | November 2017 | Part 3 | Question: 35

A text is made up of the characters a, b, c, d, e each occurring with the probability 0.11, 0.40, 0.16, 0.09 and 0.24 respectively. The optimal Huffman coding technique will have the average length of

- A. 2.40 B. 2.16 C. 2.26 D. 2.15

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Answer key

41.0.36 UGC NET CSE | November 2017 | Part 3 | Question: 36

An undirected graph $G(V, E)$ contains $n(n > 2)$ nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i and v_j are connected if and only if $0 < |i - j| \leq 2$. Each edge (v_i, v_j) is assigned a weight $i + j$. The cost of the minimum spanning tree of such a graph with 10 nodes is

- A. 88 B. 91 C. 49 D. 21

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Answer key

41.0.37 UGC NET CSE | November 2017 | Part 3 | Question: 37

An XML document that adheres to syntax rules specified by XML 1.0 specification in that it must satisfy both physical and logical structured, is called

- A. Well-formed B. Reasonable C. Valid D. Sophisticated

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Answer key

41.0.38 UGC NET CSE | November 2017 | Part 3 | Question: 38

Which of the following statement(s) is/are TRUE regarding Java Servlets?

- i. A Java Servlet is a server-side component that runs on the web server and extends the capabilities of a server.
- ii. A servlet can use the user interface classes like AWT or Swing.

- A. Only (i) is TRUE B. Only (ii) is TRUE
C. Both (i) and (ii) are TRUE D. Neither (i) nor (ii) is TRUE

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Answer key

41.0.39 UGC NET CSE | November 2017 | Part 3 | Question: 39

Consider the following HTML table definition:

```
<table border =1>
  <tr>
    <td colspan=2> Text A </td>
  </tr>
  <tr>
    <td> Text B </td>
    <td>zText C </td>
  </tr>
  <tr>
    <td rowspan =2> Text D </td>
    <td> Text E </td>
  </tr>
  <tr>
    <td> Text F </td>
  </tr>
</table>
```

The above HTML code will render on screen as

Text A	
Text B	Text C
Text D	Text E
Text F	

A.

Text A	Text B
Text C	
Text D	Text E
	Text F

B.

Text A	
Text B	Text C
Text D	Text E
	Text F

C.

Text A	
Text B	Text C
Text D	Text E
Text F	

D.

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Answer key 

41.0.40 UGC NET CSE | November 2017 | Part 3 | Question: 41



Which of the following statements is/are TRUE regarding JAVA?

- i. Constants that cannot be changed are declared using 'static' keyword.
- ii. A class can only inherit one class but can implement multiple interfaces.

A. Only (i) is TRUE

B. Only (ii) is TRUE

C. Both (i) and (ii) are TRUE

D. Neither (i) nor (ii) is TRUE

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Answer key 

41.0.41 UGC NET CSE | November 2017 | Part 3 | Question: 42



What is the output of the following JAVA program?

```

Class Test
{
    public static void main (String [] args)
    {
        Test obj=new Test ();
        obj.start ();
    }
    void start()
    {
        String stra="do";
        String strb=method (stra);
        System.out.print(": " +stra +strb);
    }
    String method (String stra)
    {
        stra=stra+"good";
        System.out.print(stra);
        return "good";
    }
}

```

A. dogood : dogoodgood

B. dogood : gooddogood

C. dogood : dodogood

D. dogood : dogood

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Answer key 

41.0.42 UGC NET CSE | November 2017 | Part 3 | Question: 40

Which of the following statements is/are TRUE?

- i. In HTML, character entities are used to incorporate external content into a web page, such as images.
- ii. Once a web server returns a cookie to a browser, the cookie will be included in all future requests from the browser to the same server.

- A. Only (i) is TRUE
- B. Only (ii) is TRUE
- C. Both (i) and (ii) are TRUE
- D. Neither (i) nor (ii) is TRUE

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41.0.43 UGC NET CSE | November 2017 | Part 3 | Question: 47

A simple stand – alone software utility is to be developed in ‘C’ programming by a team of software experts for a computer running Linux and the overall size of this software is estimated to be 20,000 lines of code. Considering $(a, b) = (2.4, 1.05)$ as multiplicative and exponentiation factor for the basic COCOMO effort estimation equation and $(c, d) = (2.5, 0.38)$ as multiplicative and exponentiation factor for the basic COCOMO development time estimation equation, approximately how long does the software project take to complete?

- A. 10.52 months
- B. 11.52 months
- C. 12.52 months
- D. 14.52 months

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Answer key

41.0.44 UGC NET CSE | November 2017 | Part 3 | Question: 43

Statistical software quality assurance in software engineering involves _____

- A. using sampling in place of exhaustive testing of software
- B. surveying customers to find out their opinions about product quality
- C. tracing each defect to its underlying cause, isolating the vital few causes, and moving to correct them
- D. tracing each defect to its underlying causes, and using the Pareto principle to correct each problem found

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Answer key

41.0.45 UGC NET CSE | November 2017 | Part 3 | Question: 46

Consider the method mcq ();

```
int mcq (boolean a, boolean b, boolean c, boolean d)
{
    int ans=1;
    if (a) {ans=2;}
    else if (b) {ans=3;}
    else if (c) {
        if (d) {ans=4;}
    }
    return ans;
}
```

If

$M1$ = Number of tests to exhaustively test mcq();

$M2$ = Minimum number of tests to achieve full statement coverage for mcq (); and

$M3$ = Minimum number of tests to achieve full branch coverage for mcq ();

then $(M1, M2, M3) =$ _____

- A. (16,3,5)
- B. (8,5,3)
- C. (8,3,5)
- D. (16,4,4)

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Answer key

41.0.46 UGC NET CSE | November 2017 | Part 3 | Question: 44

Which of the following statements is/are FALSE with respect to software testing?

S1: White-box tests are based on specifications; better at telling whether program meets specification, better at finding errors of omission.

S2: Black-box tests are based on code; better for finding crashes, out of bounds errors, file not closed errors.

S3: Alpha testing is conducted at the developer's site by a team of highly skilled testers for software that is developed as a product to be used by many customers.

- A. Only *S1* and *S2* are FALSE
C. Only *S2* and *S3* are FALSE

- B. Only *S1* and *S3* are FALSE
D. All of *S1*, *S2*, and *S3* are FALSE

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41.0.47 UGC NET CSE | November 2017 | Part 3 | Question: 45

A signal processor software is expected to operate for 91.25 days after repair, and the mean software repair time is expected to be 5 minutes. Then, the availability of the software is

- A. 96.9862% B. 97.9862% C. 98.9962% D. 99.9962%

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41.0.48 UGC NET CSE | November 2017 | Part 3 | Question: 49

Consider the following four processes with the arrival time and length of CPU burst given in milliseconds:

Process	Arrival Time	Burst Time
P_1	0	8
P_2	1	4
P_3	2	9
P_4	3	5

The average waiting time for preemptive SJF scheduling algorithm is _____

- A. 6.5 ms B. 7.5 ms C. 6.75 ms D. 7.75 ms

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Answer key

41.0.49 UGC NET CSE | November 2017 | Part 3 | Question: 48

In Software Configuration Management (SCM), which of the following is a use-case supported by standard version control systems?

- a. Managing several versions or release of a software b. Filing bug reports and tracking their progress
c. Allowing team members to work in parallel d. Identifying when and where a regression occurred
A. Only (a), (c), and (d) B. Only (a), (b), and (c)
C. Only (a), (b), and (d) D. Only (b), (c), and (d)

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Answer key

41.0.50 UGC NET CSE | November 2017 | Part 3 | Question: 51

User level threads are threads that are visible to the programmer and are unknown to the kernel. The operating system kernel supports and manages kernel level threads. Three different types of models relate user and kernel level threads.

Which of the following statements is/are true?

- i.
i. The Many – to – one model maps many user threads to one kernel thread
ii. The one – to – one model maps one user thread to one kernel thread
iii. The many – to – many model maps many user threads to smaller or equal kernel threads

ii.

- i. Many – to – one model maps many kernel threads to one user thread
- ii. One – to – one model maps one kernel thread to one user thread
- iii. Many – to – many model maps many kernel threads to smaller or equal user threads

- A. (i) is true; (ii) is false
C. Both (i) and (ii) are true

- B. (i) is false; (ii) is true
D. Both (i) and (ii) are false

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Answer key 

41.0.51 UGC NET CSE | November 2017 | Part 3 | Question: 50



Consider a virtual page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. Suppose a demand paged virtual memory system running on a computer system such that the main memory has 3 page frames. Then _____ page replacement algorithm has minimum number of page faults.

- A. FIFO B. LIFO C. LRU D. Optimal

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Answer key 

41.0.52 UGC NET CSE | November 2017 | Part 3 | Question: 54



The Bounded buffer problem is also known as _____

- A. Producer – consumer problem B. Reader – writer problem
C. Dining Philosophers problem D. Both (2) and (3)

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Answer key 

41.0.53 UGC NET CSE | November 2017 | Part 3 | Question: 56



If b is the branching factor and m is the maximum depth of the search tree, what is the space complexity of greedy search?

- A. $O(b+m)$ B. $O(bm)$ C. $O(b^m)$ D. $O(m^b)$

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Answer key 

41.0.54 UGC NET CSE | November 2017 | Part 3 | Question: 53



Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 37, 122, 14, 124, 65, 67. Suppose SSTF disk scheduling algorithm implemented to meet the requests then the total number of head movements are _____ if the disk head is initially at 53.

- A. 224 B. 248 C. 236 D. 240

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41.0.55 UGC NET CSE | November 2017 | Part 3 | Question: 55



In Artificial Intelligence (AI), which agent deals with happy and unhappy state?

- A. Simple reflex agent B. Model based agent
C. Learning agent D. Utility based agent

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41.0.56 UGC NET CSE | November 2017 | Part 3 | Question: 57



Let P, Q, R and S be Propositions. Assume that the equivalences $P \Leftrightarrow (Q \vee \neg Q)$ and $Q \Leftrightarrow R$ hold. Then the truth value of the formula $(P \wedge Q) \Rightarrow ((P \wedge R) \vee S)$ is always

- A. True
C. Same as truth table of Q

- B. False
D. Same as truth table of S

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Answer key 

41.0.57 UGC NET CSE | November 2017 | Part 3 | Question: 58



“If X , then Y unless Z ” is represented by which of the following formulae in propositional logic?

- A. $(X \wedge Y) \rightarrow \neg Z$
C. $X \rightarrow (Y \wedge \neg Z)$
- B. $(X \wedge \neg Z) \rightarrow Y$
D. $Y \rightarrow (X \wedge \neg Z)$

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Answer key 

41.0.58 UGC NET CSE | November 2017 | Part 3 | Question: 59



Consider the following two well-formed formulas in prepositional logic.

$$F1 : P \Rightarrow \neg P$$

$$F2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$$

Which of the following statements is correct?

- A. $F1$ is Satisfiable, $F2$ is valid
C. $F1$ is unsatisfiable, $F2$ is valid
- B. $F1$ is unsatisfiable, $F2$ is Satisfiable
D. $F1$ and $F2$ both are Satisfiable

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Answer key 

41.0.59 UGC NET CSE | November 2017 | Part 3 | Question: 60



Standard planning algorithms assume environment to be _____

- A. Both deterministic and fully observable
B. Neither deterministic nor fully observable
C. Deterministic but not fully observable
D. Not deterministic but fully observable

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Answer key 

41.0.60 UGC NET CSE | November 2017 | Part 3 | Question: 61



Which of the following statements is **not** correct?

- A. Every recursive language is recursively enumerable
B. $L = \{0^n 1^n 0^n \mid n = 1, 2, 3, \dots\}$ is recursively enumerable
C. Recursive languages are closed under intersection
D. Recursive languages are not closed under intersection

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41.0.61 UGC NET CSE | November 2017 | Part 3 | Question: 64



A 4×4 DFT matrix is given by

$$\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & x & -1 & y \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

$$j^2 = -1$$

Where the values of x and y are _____, _____ respectively.

- A. 1, -1 B. -1, 1 C. $-j, j$ D. $j, -j$

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Answer key 

41.0.62 UGC NET CSE | November 2017 | Part 3 | Question: 62



Context free grammar is not closed under

- A. Concatenation B. Complementation C. Kleene Star D. Union

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Answer key 

41.0.63 UGC NET CSE | November 2017 | Part 3 | Question: 63



Consider the following languages:

$$L_1 = \{a^m b^n \mid m \neq n\}$$

$$L_2 = \{a^m b^n \mid m = 2n + 1\}$$

$$L_3 = \{a^m b^n \mid m \neq 2n\}$$

Which one of the following statements is correct?

- A. Only L_1 and L_2 are context free languages
B. Only L_1 and L_3 are context free languages
C. Only L_2 and L_3 are context free languages
D. L_1 , L_2 and L_3 are context free languages

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41.0.64 UGC NET CSE | November 2017 | Part 3 | Question: 65



Entropy of a discrete random variable with possible values $\{x_1, x_2, \dots, x_n\}$ and probability density function $P(X)$ is

$$H(X) = -\sum_{i=1}^n P(x_i) \log_b P(x_i)$$

The value of b gives the units of entropy. The unit for $b = 10$ is

- A. bits B. bann C. nats D. deca

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41.0.65 UGC NET CSE | November 2017 | Part 3 | Question: 66



For any binary (n, h) linear code with minimum distance $(2t + 1)$ or greater $n - h \geq \log_2 \left[\sum_{i=0}^{\alpha} \binom{n}{i} \right]$ where α is

- A. $2t + 1$ B. $t + 1$ C. $t - 1$ D. t

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41.0.66 UGC NET CSE | November 2017 | Part 3 | Question: 69



Consider the following LPP:

$$\text{Min } Z = 2x_1 + x_2 + 3x_3$$

Subject to:

$$x_1 - 2x_2 + x_3 \geq 4$$

$$2x_1 + x_2 + x_3 \leq 8$$

$$x_1 - x_3 \geq 0$$

$$x_1, x_2, x_3 \geq 0$$

The solution of this LPP using Dual Simplex Method is

A. $x_1 = 0, x_2 = 0, x_3 = 3$ and $Z = 9$

B. $x_1 = 0, x_2 = 6, x_3 = 0$ and $Z = 6$

C. $x_1 = 4, x_2 = 0, x_3 = 0$ and $Z = 8$

D. $x_1 = 2, x_2 = 0, x_3 = 2$ and $Z = 10$

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Answer key 

41.0.67 UGC NET CSE | November 2017 | Part 3 | Question: 67



Which of the following is a valid reason for causing degeneracy in a transportation problem? Here m is number of rows and n is number of columns in transportation table.

A. When the number of allocations is $m + n - 1$

B. When two or more occupied cells become unoccupied simultaneously

C. When the number of allocations is less than $m + n - 1$

D. When a loop cannot be drawn without using unoccupied cells, except the starting cell of the loop

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Answer key 

41.0.68 UGC NET CSE | November 2017 | Part 3 | Question: 68



Consider the following LPP:

$$\text{Max } Z = 15x_1 + 10x_2$$

Subject to the constraints

$$4x_1 + 6x_2 \leq 360$$

$$3x_1 + 0x_2 \leq 180$$

$$0x_1 + 5x_2 \leq 200$$

$$x_1, x_2 \geq 0$$

The solution of the LPP using Graphical solution technique is

A. $x_1 = 60, x_2 = 0$ and $Z = 900$

B. $x_1 = 60, x_2 = 20$ and $Z = 1100$

C. $x_1 = 60, x_2 = 30$ and $Z = 1200$

D. $x_1 = 50, x_2 = 40$ and $Z = 1150$

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Answer key 

41.0.69 UGC NET CSE | November 2017 | Part 3 | Question: 70



Consider a Takagi-Sugeno – Kang (TSK) Model consisting of rules of the form:

If x_1 is A_{i1} and $\dots$ and x_r is A_{ir}

THEN $y = f_i(x_1, x_2, \dots, x_r) = b_{i0} + b_{i1}x_1 + \dots + b_{ir}x_r$

assume, α_i is the matching degree of rule i , then the total output of the model is given by:

A. $y = \sum_{i=1}^L \alpha_i f_i(x_1, x_2, \dots, x_r)$

B. $y = \frac{\sum_{i=1}^L \alpha_i f_i(x_1, x_2, \dots, x_r)}{\sum_{i=1}^L \alpha_i}$

$$C. y = \frac{\sum_{i=1}^L f_i(x_1, x_2, \dots, x_r)}{\sum_{i=1}^L \alpha_i}$$

$$D. y = \max_i [\alpha_i f_i(x_1, x_2, \dots, x_r)]$$

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Answer key 

41.0.70 UGC NET CSE | November 2017 | Part 3 | Question: 71



Consider a single perceptron with sign activation function. The perceptron is represented by weight vector $[0.4 \quad -0.3 \quad 0.1]^t$ and a bias $\theta = 0$. If the input vector to the perceptron is $X = [0.2 \quad 0.6 \quad 0.5]$ then the output of the perceptron is

- A. 1 B. 0 C. -0.05 D. -1

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Answer key 

41.0.71 UGC NET CSE | November 2017 | Part 3 | Question: 72



The Sigmoid activation function $f(t)$ is defined as

- A. $\frac{1}{\exp(t) + \exp(-t)}$ B. $t \exp(-t)$
C. $\frac{1}{1 + \exp(t)}$ D. $\frac{1}{1 + \exp(-t)}$

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Answer key 

41.0.72 UGC NET CSE | November 2017 | Part 3 | Question: 73



Consider the following statements:

- i. UNIX provides three types of permissions
 - Read
 - Write
 - Execute
- ii. UNIX provides three sets of permissions
 - permission for owner
 - permission for group
 - permission for others

Which of the above statement/s is/are **true**?

- A. Only (i) B. Only (ii)
C. Both (i) and (ii) D. Neither (i) nor (ii)

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Answer key 

41.0.73 UGC NET CSE | November 2017 | Part 3 | Question: 74



Which of the following routing technique / techniques is/are used in distributed systems?

- a. Fixed Routing b. Virtual Routing c. Dynamic Routing
A. (a) only B. (a) and (b) only C. (c) only D. All (a), (b), (c)

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Answer key 

41.0.74 UGC NET CSE | December 2018 | Part 2 | Question: 99



Consider the sentence below:

“There is a country that borders both India and Nepal”

Which of the following represents the above sentence correctly?

- A. $\exists c \text{ Country}(c) \wedge \text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Nepal})$
- B. $\exists c \text{ Country}(c) \Rightarrow [\text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Nepal})]$
- C. $[\exists c \text{ Country}(c)] \Rightarrow [\text{Border}(c, \text{India}) \wedge \text{Border}(c, \text{Nepal})]$
- D. $\exists c \text{ Border}(\text{Country}(c), \text{India} \wedge \text{Nepal})$

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Answer key 

41.0.75 UGC NET CSE | December 2018 | Part 2 | Question: 98



Consider a vocabulary with only four propositions A, B, C and D. How many models are there for the following sentence?

$$\neg A \vee \neg B \vee \neg C \vee \neg D$$

- A. 7
- B. 8
- C. 15
- D. 16

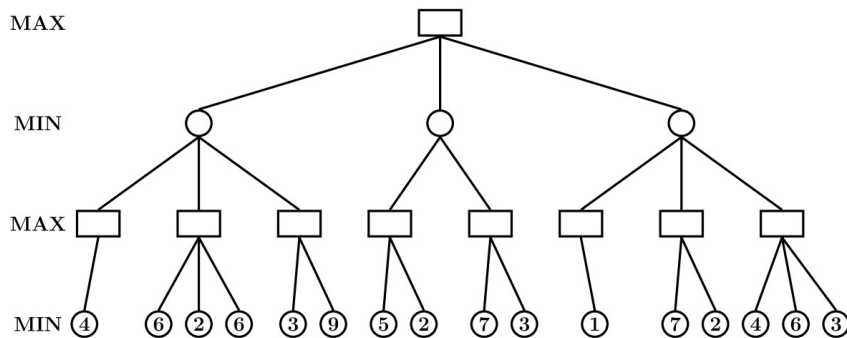
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Answer key 

41.0.76 UGC NET CSE | December 2018 | Part 2 | Question: 97



Consider the following minimax game tree search



What will be the value propagated at the root?

- A. 3
- B. 4
- C. 5
- D. 6

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Answer key 

41.0.77 UGC NET CSE | December 2018 | Part 2 | Question: 96



Consider the following statements related to *AND – OR* Search algorithm.

S1 : A solution is a subtree that has a goal node at every leaf.

S2 : *OR* nodes are analogous to the branching in a deterministic environment.

S3 : *AND* nodes are analogous to the branching in a non-deterministic environment.

Which of the following is true referencing the above statements ?

Choose the correct answer from the code given below :

Code :

- A. $S1-False, S2-True, S3-True$
- B. $S1-True, S2-True, S3-False$
- C. $S1-True, S2-True, S3-True$
- D. $S1-False, S2-True, S3-False$

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Answer key 

41.0.78 UGC NET CSE | December 2018 | Part 2 | Question: 95



Consider the following statements :

S1: A heuristic is admissible if it never overestimates the cost to reach the goal.

S2: A heuristic is monotonous if it follows triangle inequality property.

Which of the following is true referencing the above statements ?

Choose the correct answer from the code given below :

Code :

- A. Neither of the statements S1 and S2 are true
- B. Statement S1 is false but statement S2 is true
- C. Statement S1 is true but statement S2 is false
- D. Both the statements S1 and S2 are true

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Answer key 

41.0.79 UGC NET CSE | December 2018 | Part 2 | Question: 94



Match List I with List II and choose the correct answer from the code given below.

List I	List II
(a) Greedy Best-First Search	(i) Selects a node for expansion if optimal path to that node has been found
(b) A* Search	(ii) Avoids substantial overhead associated with keeping the sorted queue of nodes
(c) Recursive Best-First Search	(iii) Suffers from excessive node generation
(d) Iterative-deepening Search	(iv) Time complexity depends on the quality of heuristic

Code:

- 1. (a) – (i), (b)-(ii), (c)-(iii), (d)-(iv)
- 2. (a) – (iv), (b)-(i), (c)-(ii), (d)-(iii)
- 3. (a) – (iv), (b)-(iii), (c)-(ii), (d)-(i)
- 4. (a) – (i), (b)-(iv), (c)-(iii), (d)-(ii)

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41.0.80 UGC NET CSE | December 2018 | Part 2 | Question: 93



Consider the following terminology and match List I with List II and choose the correct answer from the code

given below.

b= branching factor

d = depth of the shallowest solution

m= maximum depth of the search tree

l=depth limit

List I (Algorithms)	List II (Space Complexity)
(a) BFS search	(i) $O(bd)$
(b) DFS search	(ii) $O(b^d)$
(c) Depth-limited search	(iii) $O(bm)$
(d) Iterative deepening search	(iv) $O(bl)$

Code:

- A. (a) – (i), (b)-(ii), (c)-(iv), (d)-(iii)
B. (a) – (ii), (b)-(iii), (c)-(iv), (d)-(i)
C. (a) – (iii), (b)-(ii), (c)-(iv), (d)-(i)
D. (a) – (i), (b)-(iii), (c)-(iv), (d)-(ii)

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Answer key 

41.0.81 UGC NET CSE | December 2018 | Part 2 | Question: 92



Which of the following is true for *semi-dynamic* environment?

- A. The environment may change while the agent is deliberating
B. The environment itself does not change with the passage of time but the agent's performance score does
C. Even if the environment changes with the passage of time while deliberating, the performance score does not change.
D. Environment and performance score, both change simultaneously

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Answer key 

41.0.82 UGC NET CSE | December 2018 | Part 2 | Question: 91



An agent can improve its performance by

- A. Perceiving B. Responding C. Learning D. Observing

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Answer key 

41.0.83 UGC NET CSE | December 2018 | Part 2 | Question: 88



Consider the following two statements:

S1 : TCP handles both congestion and flow control

S2 : UDP handles congestion but not flow control

Which of the following options is correct with respect to the above statements (S1) and (S2)?

Choose the correct answer from the code given below:

- A. Neither S1 nor S2 is correct B. S1 is not correct but S2 is correct
C. S1 is correct but S2 is not correct D. Both, S1 and S2 are correct

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Answer key 



Consider the following tables (relations):

Students	
<u>Roll-No</u>	Name
18CS101	Ramesh
18CS102	Mukesh
18CS103	Ramesh

Performance		
<u>Roll-No</u>	<u>Course</u>	Marks
18CS101	DBMS	60
18CS101	Compiler Design	65
18CS102	DBMS	80
18CS103	DBMS	85
18CS102	Compiler Design	75
18CS103	Operating System	70

Primary keys in the tables are shown using underline. Now, consider the following query:

```
SELECT S.Name, Sum (P.Marks) FROM Students, Performance P WHERE S.Roll-No=P.Roll-No GROUP BY S.Name
```

The number of rows returned by the above query is

- A. 0 B. 1 C. 2 D. 3

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Answer key



Data scrubbing is

- A. A process to reject data from the data warehouse and to create the necessary indexes.
 B. A process to load the data in the data warehouse and to create the necessary indexes.
 C. A process to upgrade the quality of data after it is moved into a data warehouse
 D. A process to upgrade the quality of data before it is moved into a data warehouse

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Answer key



Which of the following statement/s is/are true ?

- i. Facebook has the world's largest Hadoop Cluster.
 ii. Hadoop 2.0 allows live stream processing of Real time data.

Choose the correct answer from the code given below :

Code :

- A. (i) only B. (ii) only
 C. Both (i) and (ii) D. Neither (i) nor (ii)

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Answer key

41.0.87 UGC NET CSE | December 2018 | Part 2 | Question: 79



_____ command is used to remove a relation from SQL database.

- A. Drop table
- B. Delete table
- C. Remove table
- D. Update table

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Answer key

41.0.88 UGC NET CSE | December 2018 | Part 2 | Question: 78



Data warehouse contains _____ data that is never found in operational environment.

- A. Summary
- B. Encoded
- C. Encrypted
- D. Scripted

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Answer key

41.0.89 UGC NET CSE | December 2018 | Part 2 | Question: 77



An attribute A of datatype varchar(20) has the value 'xyz', and the attribute B of datatype char(20) has the value "lmnop", then the attribute A has _____ spaces and attribute B has _____ spaces.

- A. 3,5
- B. 20,20
- C. 3,20
- D. 20,5

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Answer key

41.0.90 UGC NET CSE | December 2018 | Part 2 | Question: 76



_____ system call creates new process in Unix.

- A. create
- B. create new
- C. fork
- D. fork new

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Answer key

41.0.91 UGC NET CSE | December 2018 | Part 2 | Question: 75



A process residing in Main Memory and Ready and Waiting for execution, is kept on

- A. Execution Queue
- B. Job Queue
- C. Ready Queue
- D. Wait Queue

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Answer key

41.0.92 UGC NET CSE | December 2018 | Part 2 | Question: 71



Dirty bit is used to show the

- A. wrong page
- B. page with corrupted data
- C. page with low frequency occurrence
- D. page that is modified after being loaded into cache memory

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Answer key

41.0.93 UGC NET CSE | December 2018 | Part 2 | Question: 68



In Linux operating system environment _____ command is used to print a file.

- A. print
- B. ptr
- C. pr
- D. lpr

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Answer key 

41.0.94 UGC NET CSE | December 2018 | Part 2 | Question: 67



Which of the following statements/s is/are *true* ?

- i. Windows XP supports both peer-peer and client-server networks.
- ii. Windows XP implements Transport protocols as drivers that can be loaded and unloaded from the system given dynamically.

Choose the correct answer from the code given below:

Code :

- A. (i) only
- B. (ii) only
- C. Neither (i) nor (ii)
- D. Both (i) and (ii)

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41.0.95 UGC NET CSE | December 2018 | Part 2 | Question: 65



The Third Generation mobile phones are digital and based on

- A. *AMPS*
- B. *D – AMPS*
- C. *CDMA*
- D. Broadband *CDMA*

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Answer key 

41.0.96 UGC NET CSE | December 2018 | Part 2 | Question: 62



The four byte IP address consists of

- A. Network Address
- B. Host Address
- C. Both Network and Host Addresses
- D. Neither Network nor Host Addresses

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Answer key 

41.0.97 UGC NET CSE | December 2018 | Part 2 | Question: 61



Match the following Secret Key Algorithms (list I) with Corresponding Key Lengths (List II) and choose the correct answer from the code given below:

- | List I | | List II | |
|--------|----------|---------|----------------|
| (a) | Blowfish | (i) | 128 – 256 bits |
| (b) | DES | (ii) | 128 bits |
| (c) | IDEA | (iii) | 1-448 bits |
| (d) | RC5 | (iv) | 56 bits |

Code:

- A. (a) – (iv), (b)-(ii), (c) -(ii) , d-(i)
- B. (a) – (iii), (b)-(iv), (c) -(i) , d-(ii)
- C. (a) – (iii), (b)-(iv), (c) -(ii) , d-(i)
- D. (a) – (ii), (b)-(iii), (c) -(iv) , d-(i)

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Answer key 

41.0.98 UGC NET CSE | December 2018 | Part 2 | Question: 60



Software coupling involves dependencies among pieces of software called modules. Which of the following are correct statements with respect to module coupling ?

P : Common coupling occurs when two modules share the same global data

Q : Control coupling occurs when modules share a composite data structure and use only parts of it
R : Content coupling occurs when one module modifies or relies on the internal working of another module
Choose the correct answer from the code given below :

Code :

- A. P and Q only B. P and R only C. Q and R only D. All of P, Q and R

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Answer key 

41.0.99 UGC NET CSE | December 2018 | Part 2 | Question: 59



Which of the following is **not** one of the principles of agile software development method?

- A. Customer involvement B. Embrace change
C. Incremental delivery D. Following the plan

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Answer key 

41.0.100 UGC NET CSE | December 2018 | Part 2 | Question: 58



Which of the following statements is/are true ?

P : Software Reengineering is preferable for software products having high failure rates, having poor design and/or having poor code structure.

Q : Software Reverse Engineering is the process of analyzing software with the objective of recovering its design and requirement specification.

Choose the correct answer from the code given below :

- A. *P* only B. *Q* only C. Both *P* and *Q* D. Neither *P* nor *Q*

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Answer key 

41.0.101 UGC NET CSE | December 2018 | Part 2 | Question: 56



Consider the following method:

```
int f(int m, int n, boolean x, boolean y)
{
    int res=0;
    if (m<0) {res=n-m;}
    else if (x || y)
    {
        res=-1;
        if (n==m) {res=1;}
    }
    else {res=n;}
    return res;
} /*end of f */
```

If *P* is the minimum number of tests to achieve full statement coverage for *f()*, and *Q* is the minimum number of tests to achieve full branch coverage for *f()*, then (*P*, *Q*) =

- A. (3,4) B. (4,3) C. (2,3) D. (3,2)

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Answer key 

41.0.102 UGC NET CSE | December 2018 | Part 2 | Question: 55



Which of the following statements is/are *false* ?

P : The clean-room strategy to software engineering is based on the incremental software process model.

Q : The clean-room strategy to software engineering is one of the ways to overcome “unconscious” copying of copyrighted code.

Choose the correct answer from the code given below:

Code :

- A. *P* only B. *Q* only C. Both *P* and *Q* D. Neither *P* nor *Q*

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Answer key 

41.0.103 UGC NET CSE | December 2018 | Part 2 | Question: 54



Match each UML diagram in List I to its appropriate description in List II.

- | List I | List II |
|----------------------|---|
| (a) State Diagram | (i) Describes how the external entities (people, devices) can interact with the System |
| (b) Use Case Diagram | (ii) Used to describe the static or structural view of a system |
| (c) Class Diagram | (iii) Used to show the flow of a business process, the steps of a use-case or the logic of an object behaviour |
| (d) Activity Diagram | (iv) Used to describe the dynamic behaviour of objects and could also be used to describe the entire system behaviour |

Code :

- A. (a)-(i); (b)-(iv); (c)-(ii); (d)-(iii) B. (a)-(iv); (b)-(ii); (c)-(i); (d)-(iii)
C. (a)-(i); (b)-(iv); (c)-(iii); (d)-(ii) D. (a)-(iv); (b)-(i); (c)-(ii); (d)-(iii)

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Answer key 

41.0.104 UGC NET CSE | December 2018 | Part 2 | Question: 53



Software products need perfective maintenance for which of the following reasons?

- A. To rectify bugs observed while the system is in use
B. When the customers need the product to run on new platforms
C. To support new features that users want it to support
D. To overcome wear and tear caused by the repeated use of the software

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Answer key 

41.0.105 UGC NET CSE | December 2018 | Part 2 | Question: 52



The Software Requirement Specification (*SRS*) is said to be _____ if and only if no subset of individual requirements described in it conflict with each other.

- A. Correct B. Consistent C. Unambiguous D. Verifiable

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Answer key 

41.0.106 UGC NET CSE | December 2018 | Part 2 | Question: 51



Consider the *C/C++* function *f()* given below:

```
void f(char w[])
{
    int x=strlen(w); //length of a string
```



```

char c;
for (int i=0; i<x; i++)
{
    c=w[i];
    w[i]=w[x-i-1];
    w[x-i-1] =c;
}

```

Which of the following is the purpose of $f()$?

- A. It outputs the contents of the array in reverse order
- B. It outputs the contents of the array in the original order
- C. It outputs the contents of the array with the characters shifted over by one position
- D. It outputs the contents of the array with the characters rearranged so they are no longer recognized as the words in the original phrase

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Answer key 

41.0.107 UGC NET CSE | December 2018 | Part 2 | Question: 50



Which of the following HTML5 codes will affect the horizontal as well as vertical alignment of the table content ?

- A. `<td halign="middle" valign="center"> BASH</td>`
- B. `<td align="middle" valign="center"> BASH</td>`
- C. `<td style="horizontal-align:center; vertical-align:middle;"> BASH</td>`
- D. `<td style="text-align:center; vertical-align:middle;"> BASH</td>`

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Answer key 

41.0.108 UGC NET CSE | December 2018 | Part 2 | Question: 49



What does the following Java function perform? (Assume int occupies four bytes of storage)

```

public static int f(int a)
{
    // Pre-conditions : a > 0 and no overflow/underflow occurs
    int b=0;
    for (int i=0; i<32; i++)
    {
        b = b<<1;
        b=b | ( a & 1);
        a=a >>>1; // This is a logical shift
    }
    return b;
}

```

- A. Returns the int that has the binary representation of integer a
- B. Return the int that has reversed binary representation of integer a
- C. Return the int that represents the number of 1's in the binary representation of integer a
- D. Return the int that represents the number of 0's in the binary representation of integer a

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Answer key 

41.0.109 UGC NET CSE | December 2018 | Part 2 | Question: 48



Consider the following recursive Java function f that takes two long arguments and returns a float value :

```

public static float f(long m, long n)
{
    float result = (float) m / (float) n;
    if (m<0 || n <0)

```

```

        return 0.0f;
    else
        result += f(m*2, n*3);
    return result;
}

```

Which of the following integers best approximates the value of $f(2, 3)$?

- A. 0 B. 1 C. 2 D. 3

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41.0.110 UGC NET CSE | December 2018 | Part 2 | Question: 47



Consider the following two $C++$ programs $P1$ and $P2$ and two statements $S1$ and $S2$ about these programs:

$P1$	$P2$
<pre> void f(int a, int *b, int &c) { a=1; *b=2; c=3; } int main() { int i=0; f(i, &i, i); count << i; } </pre>	<pre> double a=1, b=2; double &f(double &d) { d=4; return b; } int main() { f(a)=5; count << a << ":" << b; } </pre>

$S1$: $P1$ prints out 3.

$S2$: $P2$ prints out 4 : 2

What can you say about the statements $S1$ and $S2$?

Code :

- A. Only $S1$ is true B. Only $S2$ is true
C. Both $S1$ and $S2$ are true D. Neither $S1$ nor $S2$ is true

ugcnetcse-dec2018-paper2

Answer key

41.0.111 UGC NET CSE | December 2018 | Part 2 | Question: 46



Consider the following pseudo-code fragment, where m is a non-negative integer that has been initialized:

```

p = 0
k = 0
while (k < m)
p = p + 2k
k = k + 1
end while

```

Which of the following is a loop invariant for the while statement?

(Note: a loop invariant for a while statement is an assertion that is true each time the guard is evaluated during the execution of the while statement).

- A. $p = 2^k - 1$ and $0 \leq k < m$ B. $p = 2^{k+1} - 1$ and $0 \leq k < m$
C. $p = 2^k - 1$ and $0 \leq k \leq m$ D. $p = 2^{k+1} - 1$ and $0 \leq k \leq m$

Answer key

41.0.112 UGC NET CSE | December 2018 | Part 2 | Question: 45



In 3D Graphics, which of the following statements is/are true?

P: Back-face culling is an example of image-precision visible-surface determination procedure.

Q: Z- buffer is a 16-bit, 32-bit, or 64 bit field associated with each pixel in a frame buffer that can be used to determine the visible surfaces at each pixel

Choose the correct answer from the code given below :

Code :

- A. P only B. Q only C. P and Q D. Neither P nor Q

ugcnetcse-dec2018-paper2

Answer key

41.0.113 UGC NET CSE | December 2018 | Part 2 | Question: 44



In 3D Graphics, which of the following statements about perspective and parallel projection is/are true?

P : In a perspective projection, the farther an object is from the center of projection, the smaller it appears

Q : Parallel projection is equivalent to a perspective projection where the viewer is standing infinitely far away

R : Perspective projections do not preserve straight lines.

Choose the correct answer from the code given below:

Code :

- A. P and Q only B. P and R only C. Q and R only D. P, Q and R

ugcnetcse-dec2018-paper2

Answer key

41.0.114 UGC NET CSE | December 2018 | Part 2 | Question: 43



Consider the midpoint (or Bresenham) algorithm for rasterizing lines given below:

1. Input (x_1, y_1) and (x_2, y_2)
2. $y = y_1$
3. $d = f(x_1 + 1, y_1 + 1/2)$ // f is the implicit form of a line
4. for $x = x_1$ to x_2
5. do
6. plot(x, y)
7. if ($d < 0$)
8. then
9. $y = y + 1$
10. $d = d + (y_1 - y_2) + (x_2 - x_1)$
11. else
12. $d = d + (y_1 - y_2)$
13. end
14. end

Which statements are true?

P: For a line with slope $m > 1$, we should change the outer loop in line (4) to be over y

Q: Lines (10) and (12) update the decision variable d through an incremental evaluation of the line equation f

R: The algorithm fails if d is over 0

Choose the correct answer from the code given below:

- A. P only B. P and Q only C. Q and R only D. P, Q and R

41.0.115 UGC NET CSE | December 2018 | Part 2 | Question: 41



If the frame buffer has 10-bits per pixel and 8-bits are allocated for each of the R , G , and B components, then what would be the size of the color lookup table (LUT)?

- A. $(2^8 + 2^9)$ bytes
 B. $(2^{10} + 2^8)$ bytes
 C. $(2^{10} + 2^{24})$ bytes
 D. $(2^{10} + 2^{11})$ bytes

ugcnetcse-dec2018-paper2

Answer key

41.0.116 UGC NET CSE | December 2018 | Part 2 | Question: 40



The grammar $S \rightarrow (S) \mid SS \mid \epsilon$ is **not** suitable for predictive parsing because the grammar is

- A. Right recursive
 B. Left recursive
 C. Ambiguous
 D. An operator grammar

ugcnetcse-dec2018-paper2

Answer key

41.0.117 UGC NET CSE | December 2018 | Part 2 | Question: 38



Which one of the following problems is decidable for recursive languages (L) ?

- A. Is $L = \phi$?
 B. Is $w \in L$, where w is a string ?
 C. Is $L = \Sigma^*$?
 D. Is $L = R$, where R is a given regular set ?

ugcnetcse-dec2018-paper2

Answer key

41.0.118 UGC NET CSE | December 2018 | Part 2 | Question: 31



Let $r = a(a + b)^*$, $s = aa^*b$ and $t = a^*b$ be three regular expressions.

Consider the following :

- i. $L(s) \subseteq L(r)$ and $L(s) \subseteq L(t)$
 ii. $L(r) \subseteq L(s)$ and $L(s) \subseteq L(t)$

Choose the correct answer from the code given below :

- A. Only i is correct
 B. Only ii is correct
 C. Both i and ii are correct
 D. Neither i nor ii is correct

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Answer key

41.0.119 UGC NET CSE | December 2018 | Part 2 | Question: 28



In K-coloring of an undirected graph $G = (V, E)$ is a function. $c : V \rightarrow \{0, 1, \dots, K - 1\}$ such that $c(u) \neq c(v)$ for every edge $(u, v) \in E$.

Which of the following is **not** correct?

- A. G is bipartite
 B. G is 2-colorable
 C. G has cycles of odd length
 D. G has no cycles of odd length

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41.0.120 UGC NET CSE | December 2018 | Part 2 | Question: 26



In a ternary tree, the number of internal nodes of degree 1, 2, and 3 is 4, 3, and 3 respectively. The number of leaf nodes in the ternary tree is

- A. 9
 B. 10
 C. 11
 D. 12

Answer key **41.0.121 UGC NET CSE | December 2018 | Part 2 | Question: 20**

If a graph (G) has no loops or parallel edges, and if the number of vertices (n) the graph is $n \geq 3$, then graph G is Hamiltonian if

- i. $\deg(v) \geq \frac{n}{3}$ for each vertex v
- ii. $\deg(v) + \deg(w) \geq n$ whenever v and w are not connected by an edge
- iii. $E(G) \geq \frac{1}{3}(n-1)(n-2) + 2$

Choose the correct answer from the code given below:

- A. (i) and (iii) only B. (ii) only C. (ii) and (iii) only D. (iii) only

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Answer key **41.0.122 UGC NET CSE | December 2018 | Part 2 | Question: 19**

Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 0.5 ns, 5 ns and 100 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.7 and 0.8 respectively. What is the average access time of the system ignoring the search time within the cache?

- A. 35.20 ns B. 7.55 ns C. 20.75 ns D. 24.35 ns

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Answer key **41.0.123 UGC NET CSE | December 2018 | Part 2 | Question: 14**

A computer uses a memory unit with 256 K words of 32 bits each. A binary instruction code is stored in one word of memory. The instruction has four parts: an indirect bit, an operation code and a register code part to specify one of 64 registers and an address part. How many bits are there in the operation code, the register code part and the address part?

- A. 7,6,18 B. 6,7,18 C. 7,7,18 D. 18,7,7

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Answer key **41.0.124 UGC NET CSE | December 2018 | Part 2 | Question: 13**

Consider the following statements :

- i. Auto increment addressing mode is useful in creating self-relocating code.
- ii. If auto increment addressing mode is included in an instruction set architecture, then an additional ALU is required for effective address calculation.
- iii. In auto incrementing addressing mode, the amount of increment depends on the size of the data item accessed.

Which of the above statements is/are true?

- A. (i) and (ii) only B. (ii) and (iii) only C. (iii) only D. (ii) only

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Answer key **41.0.125 UGC NET CSE | December 2018 | Part 2 | Question: 9**

Use Dual Simplex Method to solve the following problem:

Maximize $z = -2x_1 - 3x_2$

subject to:

$$x_1 + x_2 \geq 2$$

$$2x_1 + x_2 \leq 10$$

$$x_2 + x_2 \leq 8$$

$$x_1, x_2 \geq 0$$

A. $x_1 = 2, x_2 = 0$, and $z = -4$

C. $x_1 = 0, x_2 = 2$, and $z = -6$

B. $x_1 = 2, x_2 = 6$, and $z = -22$

D. $x_1 = 6, x_2 = 2$, and $z = -18$

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41.0.126 UGC NET CSE | December 2018 | Part 2 | Question: 6



The relation $\leq$ and $>$ on a boolean algebra are defined as:

$x \leq y$ if and only if $x \vee y = y$

$x < y$ means $x \leq y$ but $x \neq y$

$x \geq y$ means $y \leq x$ and

$x > y$ means $y < x$

Considering the above definitions, which of the following is not true in the boolean algebra?

i. If $x \leq y$ and $y \leq z$, then $x \leq z$

ii. If $x \leq y$ and $y \leq x$, then $x = y$

iii. If $x < y$ and $y < z$, then $x \leq y$

iv. If $x < y$ and $y < z$, then $x < y$

Choose the correct answer from the code given below:

A. i and ii only

B. ii and iii only

C. iii only

D. iv only

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Answer key

41.0.127 UGC NET CSE | December 2018 | Part 2 | Question: 4



A survey has been conducted on methods of commuter travel. Each respondent was asked to check Bus, Train and Automobile as a major method of travelling to work. More than one answer was permitted. The results reported were as follows :

Bus 30 people; Train 35 people; Automobile 100 people; Bus and Train 15 people; Bus and Automobile 15 people; Train and Automobile 20 people; and all the three methods 5 people. How many people completed the survey form?

A. 120

B. 165

C. 160

D. 115

ugcnetcse-dec2018-paper2

Answer key

41.0.128 UGC NET CSE | October 2020 | Part 1 | Question: 50



Read the passage and answer the question:

Through their work and the efforts of others, managers in all kinds of organizational settings, whether industrial, educational, governmental, or military are attempting to run their organizations with the management by objectives process as a basic underlying management concept. Management by objectives is basically a process where by the seniors and the junior managers of an enterprise jointly identify its common goals, define each individual's major areas of responsibility in terms of the results expected... use these measures as guides for operating the unit and assessing the contribution of each of its members. In some cases, this process has been successfully carried beyond the managerial level to include hourly employees. The concept rests in a philosophy of management that emphasizes an integration between external control by managers and self-control by employees. It can apply to any manager or individual no matter what level or function, and to any organization, regardless of size. The effective functioning of this system is an agreement between a manager and an employee about that employee's groups performance goals during a stated time

period. These goals can emphasise either output variables of intervening variables or some combination of both. The important thing is that goals are jointly established and agreed upon in advance. At the end of the time period, performance is reviewed in relation to accepted goals. Both the employee and the manager participate in this review.

The philosophy behind management by objectives is to

- A. effect improvement through a joint review of achievement of performance goals within a given timeframe
- B. integrate external and internal controls by managers
- C. provide managers scope to review the accepted output variables
- D. agree upon different performance goals for managers and employees

ugcnetcse-oct2020-paper1

Answer key 

41.0.129 UGC NET CSE | October 2020 | Part 1 | Question: 49



Read the passage and answer the question:

Through their work and the efforts of others, managers in all kinds of organizational settings, whether industrial, educational, governmental, or military are attempting to run their organizations with the management by objectives process as a basic underlying management concept. Management by objectives is basically a process where by the seniors and the junior managers of an enterprise jointly identify its common goals, define each individual's major areas of responsibility in terms of the results expected... use these measures as guides for operating the unit and assessing the contribution of each of its members. In some cases, this process has been successfully carried beyond the managerial level to include hourly employees. The concept rests in a philosophy of management that emphasizes an integration between external control by managers and self-control by employees. It can apply to any manager or individual no matter what level or function, and to any organization, regardless of size. The effective functioning of this system is an agreement between a manager and an employee about that employee's group performance goals during a stated time period. These goals can emphasise either output variables of intervening variables or some combination of both. The important thing is that goals are jointly established and agreed upon in advance. At the end of the time period, performance is reviewed in relation to accepted goals. Both the employee and the manager participate in this review.

In this frame of reference, how does an organisation function effectively?

- A. By including the hourly employees in the managerial positions
- B. By identifying and executing performance targets in a collaborative mode
- C. By imposing a rigid philosophy of management
- D. By providing separate work environment for both managers and workers

ugcnetcse-oct2020-paper1

Answer key 

41.0.130 UGC NET CSE | October 2020 | Part 1 | Question: 48



Read the passage and answer the question:

Through their work and the efforts of others, managers in all kinds of organizational settings, whether industrial, educational, governmental, or military are attempting to run their organizations with the management by objectives process as a basic underlying management concept. Management by objectives is basically a process where by the seniors and the junior managers of an enterprise jointly identify its common goals, define each individual's major areas of responsibility in terms of the results expected... use these measures as guides for operating the unit and assessing the contribution of each of its members. In some cases, this process has been successfully carried beyond the managerial level to include hourly employees. The concept rests in a philosophy of management that emphasizes an integration between external control by managers and self-control by employees. It can apply to any manager or individual no matter what level or function, and to any organization, regardless of size. The effective functioning of this system is an agreement between a manager and an employee about that employee's group performance goals during a stated time period. These goals can emphasise either output variables of intervening variables or some combination of both. The important thing is that goals are jointly established and agreed upon in advance. At the end of the time period, performance is reviewed in relation to accepted goals. Both the employee and the manager participate in this review.

The critical feature of the concept in this type of management is related to

- A. the assignment of responsibility to the workers
- B. distinguishing the operational role between managers and workers
- C. promoting of uniformity of work environment among all types of organisation
- D. sharing of responsibility with an eye on results

ugnetcse-oct2020-paper1

Answer key 

41.0.131 UGC NET CSE | October 2020 | Part 1 | Question: 47



Read the passage and answer the question:

Through their work and the efforts of others, managers in all kinds of organizational settings, whether industrial, educational, governmental, or military are attempting to run their organizations with the management by objectives process as a basic underlying management concept. Management by objectives is basically a process where by the seniors and the junior managers of an enterprise jointly identify its common goals, define each individual's major areas of responsibility in terms of the results expected... use these measures as guides for operating the unit and assessing the contribution of each of its members. In some cases, this process has been successfully carried beyond the managerial level to include hourly employees. The concept rests in a philosophy of management that emphasizes an integration between external control by managers and self-control by employees. It can apply to any manager or individual no matter what level or function, and to any organization, regardless of size. The effective functioning of this system is an agreement between a manager and an employee about that employee's group performance goals during a stated time period. These goals can emphasise either output variables of intervening variables or some combination of both. The important thing is that goals are jointly established and agreed upon in advance. At the end of the time period, performance is reviewed in relation to accepted goals. Both the employee and the manager participate in this review.

The requisite for management by objectives is centered on

- | | |
|------------------------------|--------------------------------|
| A. the control by seniors | B. the guidance of juniors |
| C. the participatory process | D. the organisational settings |

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Answer key 

41.0.132 UGC NET CSE | October 2020 | Part 1 | Question: 46



Read the passage and answer the question:

Through their work and the efforts of others, managers in all kinds of organizational settings, whether industrial, educational, governmental, or military are attempting to run their organizations with the management by objectives process as a basic underlying management concept. Management by objectives is basically a process where by the seniors and the junior managers of an enterprise jointly identify its common goals, define each individual's major areas of responsibility in terms of the results expected... use these measures as guides for operating the unit and assessing the contribution of each of its members. In some cases, this process has been successfully carried beyond the managerial level to include hourly employees. The concept rests in a philosophy of management that emphasizes an integration between external control by managers and self-control by employees. It can apply to any manager or individual no matter what level or function, and to any organization, regardless of size. The effective functioning of this system is an agreement between a manager and an employee about that employee's group performance goals during a stated time period. These goals can emphasise either output variables of intervening variables or some combination of both. The important thing is that goals are jointly established and agreed upon in advance. At the end of the time period, performance is reviewed in relation to accepted goals. Both the employee and the manager participate in this review.

In an MBO, organisation is managed through the identification of

- | | |
|---------------------------------|-------------------------------|
| A. competing goals | B. common objectives |
| C. its specific characteristics | D. senior and junior managers |

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Answer key 

41.0.133 UGC NET CSE | October 2020 | Part 1 | Question: 45



What are the advantages of Choice Based Credit System (CBCS) introduced by the UGC for higher education institutions?

- i. Focuses on student centric education
- ii. Allows students to choose inter-disciplinary and intra-disciplinary courses
- iii. Makes education at par with global standards
- iv. Makes it easy to estimate the performance levels of students in terms of marks
- v. Offers flexibility for students to undertake studies at different times and at different institutions

Choose the correct answer from the options given below:

- A. (i), (ii), (iii), (iv) only
- B. (ii), (iii), (iv), (v) only
- C. (i), (ii), (iii), (v) only
- D. (iii), (iv), (v), (i) only

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Answer key 

41.0.134 UGC NET CSE | October 2020 | Part 1 | Question: 44



Given below are two statements:

- Statement I : Equality of educational opportunities is a pre-condition for achieving equity in education.
- Statement II : Equity in education can be achieved without equality of educational opportunities.

In the light of the above statements, choose the correct answer from the options given below:

- A. Both Statement *I* and Statement *II* are true
- B. Both Statement *I* and Statement *II* are false
- C. Statement *I* is correct but Statement *II* is false
- D. Statement *I* is incorrect but Statement *II* is true

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Answer key 

41.0.135 UGC NET CSE | October 2020 | Part 1 | Question: 43



Which of the following come under the mandate of the University Grants Commission (UGC)?

- i. Promotion and coordination of University education
- ii. Determining and monitoring standards of teaching, examination and research in Universities
- iii. Organising continuous professional development programmes for college and University teachers
- iv. Framing regulations on minimum standards of education
- v. Disbursing and regulating grants to the universities and colleges

Choose the correct answer from the options given below:

- A. (i), (ii), (iii), (iv) only
- B. (ii), (iii), (iv), (v) only
- C. (i), (iii), (iv), (v) only
- D. (i), (ii), (iv), (v) only

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41.0.136 UGC NET CSE | October 2020 | Part 1 | Question: 42



Which of the following explain the concept of values?

- i. Values are beliefs tied inextricably to human emotion and behaviour
- ii. Values refer to desirable goals people strive to attain
- iii. Values are conflict-free
- iv. Values are ordered by importance relative to one another
- v. Values are utilized primarily to effect change in knowledge of a person

Choose the correct answer from the options given below:

- A. (i), (ii), (iii) only
- B. (i), (ii), (iv) only
- C. (ii), (iii), (iv) only
- D. (iii), (iv), (v) only

41.0.137 UGC NET CSE | October 2020 | Part 1 | Question: 41

Match List I with List II: List I consists of the Vedangas and List II contains subjects dealt in them.

List I	List II
(A) Nirukta	(I) Phonetics
(B) Chhandas	(II) Ritual
(C) Shiksha	(III) Metrics
(D) Kalpa	(IV) Etymology

Choose the correct answer from the options given below:

- A. (A) – (II), (B) – (IV), (C) – (III), (D) – (I)
 B. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)
 C. (A) – (III), (B) – (I), (C) – (II), (D) – (IV)
 D. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)

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41.0.138 UGC NET CSE | October 2020 | Part 1 | Question: 40

Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Carbon monoxide (CO) is a serious asphyxiant; even a short exposure may have fatal health issues.

Reason (R) : Hemoglobin present in the blood has greater affinity towards carbon monoxide than oxygen

In the light of the above statements, choose the most appropriate answer from the options given below:

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
 B. Both (A) and (R) are correct but (R) is NOT the correct explanation of (A)
 C. (A) is correct but (R) is not correct
 D. (A) is not correct but (R) is correct

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Answer key

41.0.139 UGC NET CSE | October 2020 | Part 1 | Question: 39

During physical treatment process of waste water, what is the correct order of following operations to be followed?

- a. Flocculation
 c. Screening
 b. Filtration
 d. Sedimentation

Choose the correct answer from the options given below:

- A. (c), (d), (b), (a)
 B. (c), (d), (a), (b)
 C. (b), (d), (a), (c)
 D. (b), (c), (d), (a)

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41.0.140 UGC NET CSE | October 2020 | Part 1 | Question: 38

Wind energy is very sensitive to the wind velocity because wind power is directly proportional to the

- A. wind velocity
 C. square root of the wind velocity
 B. square of the wind velocity
 D. cube of the wind velocity

ugcnetcse-oct2020-paper1

41.0.141 UGC NET CSE | October 2020 | Part 1 | Question: 37

Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : Carbon dioxide emissions from bioenergy. Production have traditionally been excluded from most emission inventories and climate impact studies.

Reason (R) : Carbon dioxide emissions associated with production of bioenergy are significantly less compared to those from combustion of fossil fuels.

In the light of the above statements, choose the correct answer from the options given below:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
- C. (A) is true but (R) is false
- D. (A) is false but (R) is true

ugcnetcse-oct2020-paper1

41.0.142 UGC NET CSE | October 2020 | Part 1 | Question: 36



Which of the International Agreements/Convention/Protocols are legally binding on the member countries (parties)?

- i. Convention on Biological Diversity
- ii. Kyoto Protocol
- iii. Montreal Protocol
- iv. Paris Agreement

Choose the most appropriate answer from the options given below:

- A. *iii* only
- B. *i* and *iii* only
- C. *i, ii,* and *iii* only
- D. *ii, iii,* and *iv* only

ugcnetcse-oct2020-paper1

41.0.143 UGC NET CSE | October 2020 | Part 1 | Question: 35



Match List I with List II

List I	List II
(A) Input	(I) Mouse
(B) Output	(II) Monitor
(C) Memory	(III) Cache
(D) Software	(IV) Python

Choose the correct answer from the options given below:

- A. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
- B. (A) – (I), (B) – (III), (C) – (II), (D) – (IV)
- C. (A) – (III), (B) – (II), (C) – (I), (D) – (IV)
- D. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)

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Answer key

41.0.144 UGC NET CSE | October 2020 | Part 1 | Question: 34



The word blog is derived from the word

- A. My log
- B. We blog
- C. E blog
- D. Blogger

ugcnetcse-oct2020-paper1

Answer key

41.0.145 UGC NET CSE | October 2020 | Part 1 | Question: 33



A firewall is a software tool that protects

- i. Server
- ii. Network

- iii. Fire
- iv. Individual computer

Choose the correct answer from the options given below:

- A. *i, iii* only
- B. *ii, iii* only
- C. *i, ii, iii* only
- D. *i, ii, iv* only

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Answer key 

41.0.146 UGC NET CSE | October 2020 | Part 1 | Question: 32



Cloud computing has the following distinct characteristics:

- i. The service is hosted on the internet
- ii. It is made available by a service provider
- iii. It computes and predicts rain when the weather is cloudy
- iv. The service is fully managed by the provider

Choose the correct answer from the options given below:

- A. *i, ii* only
- B. *i, ii, iii* only
- C. *i, iv* only
- D. *i, ii, iv* only

ugcnetcse-oct2020-paper1

Answer key 

41.0.147 UGC NET CSE | October 2020 | Part 1 | Question: 30



Which one of the following pramanas has been accepted in Buddhism?

- A. Arthapatti
- B. Anupalabdhi
- C. S'abda
- D. Pratyaksha

ugcnetcse-oct2020-paper1

Answer key 

41.0.148 UGC NET CSE | October 2020 | Part 1 | Question: 31



A type of memory that holds the computer startup routine is

- A. Cache
- B. RAM
- C. DRAM
- D. ROM

ugcnetcse-oct2020-paper1

Answer key 

41.0.149 UGC NET CSE | October 2020 | Part 1 | Question: 29



Match List-I with List-II

- | List I
(Philosophical doctrines) | List II
(Philosophical schools) |
|-------------------------------------|------------------------------------|
| (A) Syadavada | (I) Nyaya |
| (B) Anupalabdhi | (II) Buddhism |
| (C) Apoha | (III) Vedanta |
| (D) Abhava | (IV) Jainism |

Choose the correct answer from the options given below:

- A. (A) – (II), (B) – (IV), (C) – (III), (D) – (I)
- B. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)
- C. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)
- D. (A) – (III), (B) – (I), (C) – (II), (D) – (IV)

ugcnetcse-oct2020-paper1

Answer key 

41.0.150 UGC NET CSE | October 2020 | Part 1 | Question: 28

Given below are two Statements:

The water of the pond is muddy. It must have rained last night.

What kind of Anumana (inference) has been used in the above statements. Choose the correct option from the following.

- i. S'esavat anumana
- ii. Comparison
- iii. Purvovat anumana
- iv. Both (A) and (B)

Choose the appropriate answer from the options given below:

- A. (i) only B. (ii) only C. (iii) only D. (iv) only

ugcnetcse-oct2020-paper1

Answer key

41.0.151 UGC NET CSE | October 2020 | Part 1 | Question: 27

Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A): Substance is always the subject of all predicates

Reason (R): Substance is self-existent and self-conceived

In the light of the above statements, choose the correct answer from the options given below:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
C. (A) is true but (R) is false
D. (A) is false but (R) is true

ugcnetcse-oct2020-paper1

41.0.152 UGC NET CSE | October 2020 | Part 1 | Question: 26

Which one of the following schools has not accepted anumana (inference) as a valid source of knowledge?

- A. Advaita Vedanta B. Visistadvaita C. Charvaka D. Sankhya

ugcnetcse-oct2020-paper1

Answer key

41.0.153 UGC NET CSE | October 2020 | Part 1 | Question: 25

Three pipes fill a tank in 4 hours. If two of them take 8 and 12 hours each to fill the tank, how many hours will the third pipe take to fill the tank?

- A. 6 hours B. 15 hours C. 18 hours D. 24 hours

ugcnetcse-oct2020-paper1

Answer key

41.0.154 UGC NET CSE | October 2020 | Part 1 | Question: 24

In a 100 meters race, A beat B by 10 meter or 2second. What is the speed of A?

- A. 5 Km/h B. 10 Km/h C. 15 Km/h D. 20 Km/h

ugcnetcse-oct2020-paper1

Answer key



If some pens are pencils but no pencil is sharpener then

- i. All pencils are pens
- ii. Some sharpeners are pencils
- iii. Some pens are not sharpener
- iv. Some pencils are not pens

Choose the correct answer from the options given below:

- A. (i) and (iii) only
- B. (ii) and (iii) only
- C. (i) and (iv) only
- D. (ii) and (iv) only

ugcnetcse-oct2020-paper1

Answer key



The ratio of ages of two persons A and B is $3 : 4$ and that of ages B and C is $4 : 5$. If the average of all three is 40 years, what is B 's age?

- A. 20 years
- B. 30 years
- C. 40 years
- D. 50 years

ugcnetcse-oct2020-paper1

Answer key



In a park, A, B, C, D, E and F are sitting. C is the brother of D 's husband and son of A . F is the daughter of D and sister of E .

- i. E is the grandson of A
- ii. F is the daughter of C
- iii. B is the father of F
- iv. There are one son and one daughter in the group
- v. C is the uncle of E

Choose the correct answer from the options given below:

- A. (i), (ii) and (iii) only
- B. (i), (iii) and (v) only
- C. (ii), (iv) and (v) only
- D. (ii), (iii) and (iv) only

ugcnetcse-oct2020-paper1

Answer key



Match List-I with List-II

List I	List II
(Medium)	(Example for description)
(A) Newspaper	(I) Networking
(B) Radio and television	(II) Audio, Video and graphics
(C) Social Media	(III) Broadsheet
(D) Multi-media	(IV) Broadcasting

Choose the correct answer from the options given below:

- A. $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$
- B. $(A) - (IV), (B) - (I), (C) - (II), (D) - (III)$
- C. $(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$
- D. $(A) - (II), (B) - (III), (C) - (IV), (D) - (I)$

ugcnetcse-oct2020-paper1

Answer key

41.0.159 UGC NET CSE | October 2020 | Part 1 | Question: 19

Identifying the sequence of stages of message processing

- i. Action
- ii. Internalisation
- iii. Selection
- iv. Understanding
- v. Perception

Choose the correct answer from the options given below

- | | |
|--------------------------------|--------------------------------|
| A. (i), (ii), (iv), (v), (iii) | B. (ii), (iii), (i), (iv), (v) |
| C. (iv), (v), (ii), (i), (iii) | D. (v), (iv), (iii), (ii), (i) |

ugcnetcse-oct2020-paper1

41.0.160 UGC NET CSE | October 2020 | Part 1 | Question: 18

Given below are two statements:

Statement I : Interactive and multimedia systems provide for a new culture of teaching and learning

Statement II : Virtual classrooms convert students into senseless humanoids

In the light of the above statements, choose the correct answer from the options given below:

- A. Both Statement *I* and Statement *II* are true
- B. Both Statement *I* and Statement *II* are false
- C. Statement *I* is correct but Statement *II* is false
- D. Statement *I* is incorrect but Statement *II* is true

ugcnetcse-oct2020-paper1

Answer key

41.0.161 UGC NET CSE | October 2020 | Part 1 | Question: 17

Which of the following are useful in positive components of classroom communication?

- i. Restrictive actions
- ii. Empathy
- iii. Punctuality
- iv. Follow-up
- v. Scattered reflections

Choose the most appropriate answer from the options given below:

- | | |
|--------------------------|---------------------------|
| A. (i), (ii), (iii) only | B. (ii), (iii), (iv) only |
| C. (iii), (iv), (v) only | D. (i), (iv), (v) only |

ugcnetcse-oct2020-paper1

41.0.162 UGC NET CSE | October 2020 | Part 1 | Question: 16

Which of the following is considered as an effective form of verbal communication?

- | | |
|-------------------------------|-----------------|
| A. Story-telling | B. Sloganeering |
| C. Use of high pitch of voice | D. Monologues |

ugcnetcse-oct2020-paper1

Answer key

41.0.163 UGC NET CSE | October 2020 | Part 1 | Question: 15

Given below are the two statements:

Statement I : Qualitative research is an umbrella term to refer to various research strategies that share certain characteristics. The data collected have been termed 'soft' i.e., rich in description

Statement II : Quantitative research is mainly hypothetico deductive and employs mostly statistical techniques for data analysis

In the light of the above statements, choose the correct answer from the options given below:

- A. Both Statement *I* and Statement *II* are true
- B. Both Statement *I* and Statement *II* are false
- C. Statement *I* is correct but Statement *II* is false
- D. Statement *I* is incorrect but Statement *II* is true

ugcnetcse-oct2020-paper1

Answer key 

41.0.164 UGC NET CSE | October 2020 | Part 1 | Question: 14



Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R)

Assertion (A) : In order to bring the gap between theory and practice both fundamental and applied research formats have to be promoted.

Reason (R) : Fundamental research focuses on theory making and applied research is directed at exploring the applicability of theory in varied practical situations

In the light of the above statements, choose the most appropriate answer from the options given below:

- A. Both (A) and (R) are correct and (R) is the correct explanation of (A)
- B. Both (A) and (R) are correct but (R) is NOT the correct explanation of (A)
- C. (A) is correct and (R) is not incorrect
- D. (A) is not correct and (R) is correct

ugcnetcse-oct2020-paper1

Answer key 

41.0.165 UGC NET CSE | October 2020 | Part 1 | Question: 13



Which of the following are non-probability sampling methods?

1. Cluster sampling
2. Judgemental sampling
3. Systematic sampling
4. Snowball sampling
5. Quota sampling

Choose the correct answer from the options given below:

- A. 1, 2 and 3 only
- B. 2, 4 and 5 only
- C. 2, 3 and 4 only
- D. 3, 4 and 5 only

ugcnetcse-oct2020-paper1

Answer key 

41.0.166 UGC NET CSE | October 2020 | Part 1 | Question: 12



In conducting an empirical study, a researcher employs a non-parametric test for data analysis and finds that the 'statistics' arrived at is 'significant' at .05 level. What decisions will be warranted thereafter?

1. Rejecting the Null hypothesis (H_0)
2. Accepting the Null hypothesis (H_0)

3. Accepting the alternate hypothesis (H_1)
4. Keeping the decision in abeyance
5. Rejecting the alternate hypothesis (H_1)

Choose the correct answer from the options given below:

- A. 1 and 2 only B. 2 and 3 only C. 4 only D. 1 and 3 only

ugcnetcse-oct2020-paper1

41.0.167 UGC NET CSE | October 2020 | Part 1 | Question: 11



In which of the following, there is a greater flexibility in both the methods and process of research?

- A. Ethnography and phenomenology B. Descriptive survey and impact studies
C. Experimental and observation based studies D. Expost facto and historical studies

ugcnetcse-oct2020-paper1

41.0.168 UGC NET CSE | October 2020 | Part 1 | Question: 10



Given below are two statements:

- Statement I : Indirect instruction is an approach to teaching and learning in which the process is inquiry, the content involves concept and the context is a problem
- Statement II : In NRT test data tell us about a student's level of proficiency in mastery of some skills or set of skills

In the light of the above statements, choose the correct answer from the options given below:

- A. Both Statement *I* and Statement *II* are true
B. Both Statement *I* and Statement *II* are false
C. Statement *I* is correct but and Statement *II* is false
D. Statement *I* is incorrect but Statement *II* is true

ugcnetcse-oct2020-paper1

41.0.169 UGC NET CSE | October 2020 | Part 1 | Question: 9



Match **List-I** with **List-II**:

- | List-I
(Scales of measurement) | List-II
(Description of properties) |
|--|---|
| (A) Nominal | (I) Classification and order |
| (B) Ordinal | (II) Classification |
| (C) Interval | (III) Classification, order and equal units |
| (D) Ratio | (IV) Classification, order, equal units and Absolute Zero |

Choose the correct answer from the options given below:

- A. (A) – (IV), (B) – (III), (C) – (II), (D) – (I) B. (A) – (II), (B) – (I), (C) – (III), (D) – (IV)
C. (A) – (I), (B) – (II), (C) – (IV), (D) – (III) D. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

ugcnetcse-oct2020-paper1

41.0.170 UGC NET CSE | October 2020 | Part 1 | Question: 8



In which level of teaching, the teacher's role is more active rather than being interactive?

- A. Memory level B. Understanding level
C. Reflective level D. Autonomous development level

ugcnetcse-oct2020-paper1

Answer key

41.0.171 UGC NET CSE | October 2020 | Part 1 | Question: 7



Identify the teaching strategies which are employed during indirect instruction:

- i. Emphasizing direct and indirect exemplars
- ii. Presenting the stimulus material
- iii. Inquiry based question-answer session
- iv. Presenting, asking and providing feed back
- v. Promoting problem solving approach

Choose the correct answer from the options given below

- A. *i, ii and iii* only
- B. *ii, iii and iv* only
- C. *iii, iv and v* only
- D. *i, iii and v* only

ugcnetcse-oct2020-paper1

41.0.172 UGC NET CSE | October 2020 | Part 1 | Question: 6



Which of the following teacher competency belongs to the domain of personality and attitude?

- A. Locus of control and self-efficacy
- B. Managing and Monitoring
- C. Planning and instructing
- D. Personalising and contextualizing

ugcnetcse-oct2020-paper1

Answer key

41.0.173 UGC NET CSE | October 2020 | Part 2 | Question: 62



Which of the following statements are true?

- i. A sentence α entails another sentence β if β is true in few worlds where α is true
- ii. Forward chaining and backward chaining are very natural reasoning algorithms for knowledge bases in Horn form
- iii. Sound inference algorithms derive all sentences that are entailed
- iv. Propositional logic does not scale to environments of unbounded size

Choose the correct answer from the options given below:

- A. *(i)* and *(ii)* only
- B. *(ii)* and *(iii)* only
- C. *(iii)* and *(iv)* only
- D. *(ii)* and *(iv)* only

ugcnetcse-oct2020-paper2

Answer key

41.1 Binary Heap (1)

41.1.1 Binary Heap: UGC NET CSE | December 2018 | Part 2 | Question: 22



The elements 42, 25, 30, 40, 22, 35, 26 are inserted one by one in the given order into a max-heap. The resultant max-heap is sorted in an array implementation as

- A. $\langle 42, 40, 35, 25, 22, 30, 26 \rangle$
- B. $\langle 42, 35, 40, 22, 25, 30, 26 \rangle$
- C. $\langle 42, 40, 35, 25, 22, 26, 30 \rangle$
- D. $\langle 42, 35, 40, 22, 25, 26, 30 \rangle$

ugcnetcse-dec2018-paper2 data-structures binary-heap

Answer key

41.2 Binary Search Tree (1)

41.2.1 Binary Search Tree: UGC NET CSE | December 2018 | Part 2 | Question: 25



A binary search tree is constructed by inserting the following numbers in order :

60, 25, 72, 15, 30, 68, 101, 13, 18, 47, 70, 34

The number of nodes in the left subtree is

A. 5

B. 6

C. 7

D. 3

ugcnetcse-dec2018-paper2 data-structures binary-search-tree

Answer key 

41.3 CO and Architecture (2)

41.3.1 CO and Architecture: UGC NET CSE | November 2017 | Part 2 | Question: 46



Which of the following Super Computers is the fastest Super Computer?

A. Sun-way TaihuLight

B. Titan

C. Piz Daint

D. Sequoia

ugcnetcse-nov2017-paper2 co-and-architecture

Answer key 

41.3.2 CO and Architecture: UGC NET CSE | November 2017 | Part 2 | Question: 50



Which speed up could be achieved according to Amdahl's Law for infinite number of processes if 5% of a program is sequential and the remaining part is ideally parallel?

A. Infinite

B. 5

C. 20

D. 50

ugcnetcse-nov2017-paper2 co-and-architecture

Answer key 

41.4 Clustering (1)

41.4.1 Clustering: UGC NET CSE | November 2017 | Part 2 | Question: 48



Which of the following is **not** a Clustering method?

A. K-Means method

B. Self Organizing feature map method

C. K- nearest neighbor method

D. Agglomerative method

ugcnetcse-nov2017-paper2 data-mining clustering

Answer key 

41.5 Database Normalization (1)

41.5.1 Database Normalization: UGC NET CSE | December 2018 | Part 2 | Question: 82



Consider a relation schema $R = (A, B, C, D, E, F)$ on which the following functional dependencies hold :

- $A \rightarrow B$
- $B, C \rightarrow D$
- $E \rightarrow C$
- $D \rightarrow A$

What are the candidate keys of R ?

A. AE and BE

B. AE, BE and DE

C. AEF, BEF and BCF

D. AEF, BEF and DEF

ugcnetcse-dec2018-paper2 databases database-normalization

Answer key 

41.6 Disk Scheduling (1)

41.6.1 Disk Scheduling: UGC NET CSE | November 2017 | Part 2 | Question: 37



In _____ disk scheduling algorithm, the disk head moves from one end to other end of the disk, serving the requests along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without serving any requests on the return trip.

A. LOOK

B. SCAN

C. C-LOOK

D. C-SCAN

ugcnetcse-nov2017-paper2 operating-system disk-scheduling

Answer key 

41.7

Dynamic Programming (1)

41.7.1 Dynamic Programming: UGC NET CSE | December 2018 | Part 2 | Question: 23



Consider two sequences X and Y :

$X = \langle 0, 1, 2, 1, 3, 0, 1 \rangle$

$Y = \langle 1, 3, 2, 0, 1, 0 \rangle$

The length of longest common subsequence between X and Y is

A. 2

B. 3

C. 4

D. 5

ugcnetcse-dec2018-paper2 algorithms dynamic-programming

Answer key 

41.8

E Technologies (1)

41.8.1 E Technologies: UGC NET CSE | November 2017 | Part 2 | Question: 47



Which of the following statements about ERP systems is **true**?

- A. Most ERP software implementations fully achieve seamless integration
- B. ERP software packages are themselves combinations of separate applications for manufacturing, materials, resource planning, general ledger, human resources, procurement and order entry
- C. Integration of ERP systems can be achieved in only one way
- D. An ERP package implemented uniformly throughout an enterprise is likely to contain very flexible connections to allow changes and software variations

ugcnetcse-nov2017-paper2 e-technologies

Answer key 

41.9

Ethernet (1)

41.9.1 Ethernet: UGC NET CSE | November 2017 | Part 2 | Question: 28



The number of bits used for addressing in Gigabit Ethernet is _____.

A. 32 bits

B. 48 bits

C. 64 bits

D. 128 bits

ugcnetcse-nov2017-paper2 computer-networks ethernet

Answer key 

41.10

Graph Algorithms (1)

41.10.1 Graph Algorithms: UGC NET CSE | December 2018 | Part 2 | Question: 27



Match List I with List II and choose the correct answer from the code given below.

List I

(Graph Algorithm)

- (a) Dijkstra's algorithm
- (b) Kruskal's algorithm
- (c) Floyd-Warshall algorithm
- (d) Topological sorting

List II

(Time Complexity)

- (i) $O(E \lg E)$
- (ii) $\Theta(V^3)$
- (iii) $O(V^2)$
- (iv) $\Theta(V + E)$

where V and E are the number of vertices and edges in graph respectively.

Code :

- A. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
C. (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

- B. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
D. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

ugcnetcse-dec2018-paper2 algorithms graph-algorithms time-complexity

Answer key 

41.11

IP Addressing (1)

41.11.1 IP Addressing: UGC NET CSE | November 2017 | Part 2 | Question: 30



The IP address _____ is used by hosts when they are being booted.

- A. 0.0.0.0 B. 1.0.0.0 C. 1.1.1.1 D. 255.255.255.255

ugcnetcse-nov2017-paper2 computer-networks ip-addressing

Answer key 

41.12

Indexing (1)

41.12.1 Indexing: UGC NET CSE | December 2018 | Part 2 | Question: 85



A clustering index is defined on the fields which are of type

- A. non-key and ordering B. non-key and non-ordering
C. key and ordering D. key and non-ordering

ugcnetcse-dec2018-paper2 databases indexing

Answer key 

41.13

Linked List (1)

41.13.1 Linked List: UGC NET CSE | December 2018 | Part 2 | Question: 29



Consider a singly linked list. What is the worst case time complexity of the best-known algorithm to delete the node a , pointer to this node is q , from the list?

- A. $O(n \lg n)$ B. $O(n)$ C. $O(\lg n)$ D. $O(1)$

ugcnetcse-dec2018-paper2 data-structures linked-list time-complexity

Answer key 

41.14

Microprocessors (1)

41.14.1 Microprocessors: UGC NET CSE | December 2015 | Part 3 | Question: 3



Which of the following 8085 microprocessor hardware interrupt has the lowest priority?

- A. RST 6.5 B. RST 7.5 C. TRAP D. INTR

ugcnetcse-dec2015-paper3 microprocessors 8085-microprocessor

Answer key 

41.15

Minimum Spanning Tree (1)

41.15.1 Minimum Spanning Tree: UGC NET CSE | December 2018 | Part 2 | Question: 12



Consider the graph shown below:

Answer key **41.20 Software Testing (1)****41.20.1 Software Testing: UGC NET CSE | November 2017 | Part 2 | Question: 45**

Which of the following testing techniques ensures that the software products runs correctly after the changes in maintenance?

- A. Path Testing
C. Unit Testing
- B. Integration Testing
D. Regression Testing

ugcnetcse-nov2017-paper2 is&software-engineering software-testing

Answer key **41.21 Sorting (1)****41.21.1 Sorting: UGC NET CSE | December 2018 | Part 2 | Question: 30**

The second smallest of n elements can be found with ____ comparisons in the worst case.

- A. $n - 1$
C. $n + \text{ceil}(\lg n) - 2$
- B. $\lg n$
D. $\frac{3n}{2}$

ugcnetcse-dec2018-paper2 algorithms sorting

41.22 Stack (1)**41.22.1 Stack: UGC NET CSE | December 2018 | Part 2 | Question: 24**

Consider the following postfix expression with single digit operands :

$6\ 2\ 3\ * / 4\ 2\ * + 6\ 8\ * -$

The top two elements of the stack after the second $*$ is evaluated, are :

- A. 8,2
B. 8,1
C. 6,2
D. 6,3

ugcnetcse-dec2018-paper2 data-structures stack

Answer key **41.23 Transaction and Concurrency (1)****41.23.1 Transaction and Concurrency: UGC NET CSE | December 2018 | Part 2 | Question: 83**

Consider the following sequence of two transactions on a bank account (A) with initial balance 20,000 that transfers 5,000 to another account (B) and then apply 10% interest.

- i. T1 start
- ii. T1 A old=20,000 new 15,000
- iii. T1 B old = 12,000 new = 17,000
- iv. T1 commit
- v. T2 start
- vi. T2 A old = 15,000 new = 16,500
- vii. T2 commit

Suppose the database system crashes just before log record (vii) is written. When the system is restarted, which one statement is true of the recovery process?

- A. We must redo log record (vi) to set A to 16,500
- B. We must redo log record (vi) to set A to 16,500 and then redo log records (ii) and (iii)
- C. We need not redo log records (ii) and (iii) because transaction $T1$ is committed
- D. We can apply redo and undo operations in arbitrary order because they are idempotent

Answer key **Answer Keys**

41.0.1	D	41.0.2	C	41.0.3	A	41.0.4	B	41.0.5	A
41.0.6	C	41.0.7	A	41.0.8	B	41.0.9	B	41.0.10	B;C;D
41.0.11	C	41.0.12	D	41.0.13	C	41.0.14	B	41.0.15	A
41.0.16	D	41.0.17	C	41.0.18	C	41.0.19	C	41.0.20	B
41.0.21	B	41.0.22	A	41.0.23	C	41.0.24	D	41.0.25	C
41.0.26	B	41.0.27	D	41.0.28	B	41.0.29	B	41.0.30	B
41.0.31	A	41.0.32	C	41.0.33	D	41.0.34	C	41.0.35	B
41.0.36	B	41.0.37	A	41.0.38	A	41.0.39	C	41.0.40	B
41.0.41	D	41.0.42	D	41.0.43	B	41.0.44	C	41.0.45	A
41.0.46	A	41.0.47	D	41.0.48	A	41.0.49	A	41.0.50	A
41.0.51	D	41.0.52	A	41.0.53	C	41.0.54	C	41.0.55	D
41.0.56	A	41.0.57	B	41.0.58	A	41.0.59	A	41.0.60	D
41.0.61	D	41.0.62	B	41.0.63	D	41.0.64	X	41.0.65	D
41.0.66	C	41.0.67	C	41.0.68	B	41.0.69	B	41.0.70	D
41.0.71	D	41.0.72	C	41.0.73	D	41.0.74	Q-Q	41.0.75	Q-Q
41.0.76	Q-Q	41.0.77	Q-Q	41.0.78	Q-Q	41.0.79	Q-Q	41.0.80	Q-Q
41.0.81	Q-Q	41.0.82	Q-Q	41.0.83	Q-Q	41.0.84	Q-Q	41.0.85	Q-Q
41.0.86	Q-Q	41.0.87	Q-Q	41.0.88	Q-Q	41.0.89	Q-Q	41.0.90	Q-Q
41.0.91	Q-Q	41.0.92	Q-Q	41.0.93	Q-Q	41.0.94	Q-Q	41.0.95	Q-Q
41.0.96	Q-Q	41.0.97	Q-Q	41.0.98	Q-Q	41.0.99	Q-Q	41.0.100	Q-Q
41.0.101	Q-Q	41.0.102	Q-Q	41.0.103	Q-Q	41.0.104	Q-Q	41.0.105	Q-Q
41.0.106	B	41.0.107	Q-Q	41.0.108	Q-Q	41.0.109	Q-Q	41.0.110	Q-Q
41.0.111	Q-Q	41.0.112	Q-Q	41.0.113	Q-Q	41.0.114	Q-Q	41.0.115	Q-Q
41.0.116	Q-Q	41.0.117	Q-Q	41.0.118	A	41.0.119	Q-Q	41.0.120	B
41.0.121	Q-Q	41.0.122	B	41.0.123	A	41.0.124	Q-Q	41.0.125	Q-Q
41.0.126	Q-Q	41.0.127	Q-Q	41.0.128	A	41.0.129	B	41.0.130	D
41.0.131	C	41.0.132	B	41.0.133	C	41.0.134	C	41.0.135	D
41.0.136	B	41.0.137	B	41.0.138	A	41.0.139	B	41.0.140	D
41.0.141	B	41.0.142	B	41.0.143	D	41.0.144	B	41.0.145	D
41.0.146	D	41.0.147	D	41.0.148	D	41.0.149	B	41.0.150	A
41.0.151	B	41.0.152	C	41.0.153	D	41.0.154	D	41.0.155	A
41.0.156	C	41.0.157	B	41.0.158	A	41.0.159	D	41.0.160	C
41.0.161	B	41.0.162	A	41.0.163	A	41.0.164	A	41.0.165	B
41.0.166	D	41.0.167	A	41.0.168	C	41.0.169	B	41.0.170	A
41.0.171	D	41.0.172	A	41.0.173	D	41.1.1	Q-Q	41.2.1	C

41.3.1	A
41.7.1	4
41.12.1	Q-Q
41.17.1	D
41.22.1	B

41.3.2	C
41.8.1	B
41.13.1	B
41.18.1	Q-Q
41.23.1	Q-Q

41.4.1	C
41.9.1	B
41.14.1	X
41.19.1	C

41.5.1	Q-Q
41.10.1	Q-Q
41.15.1	Q-Q
41.20.1	D

41.6.1	D
41.11.1	A
41.16.1	Q-Q
41.21.1	Q-Q